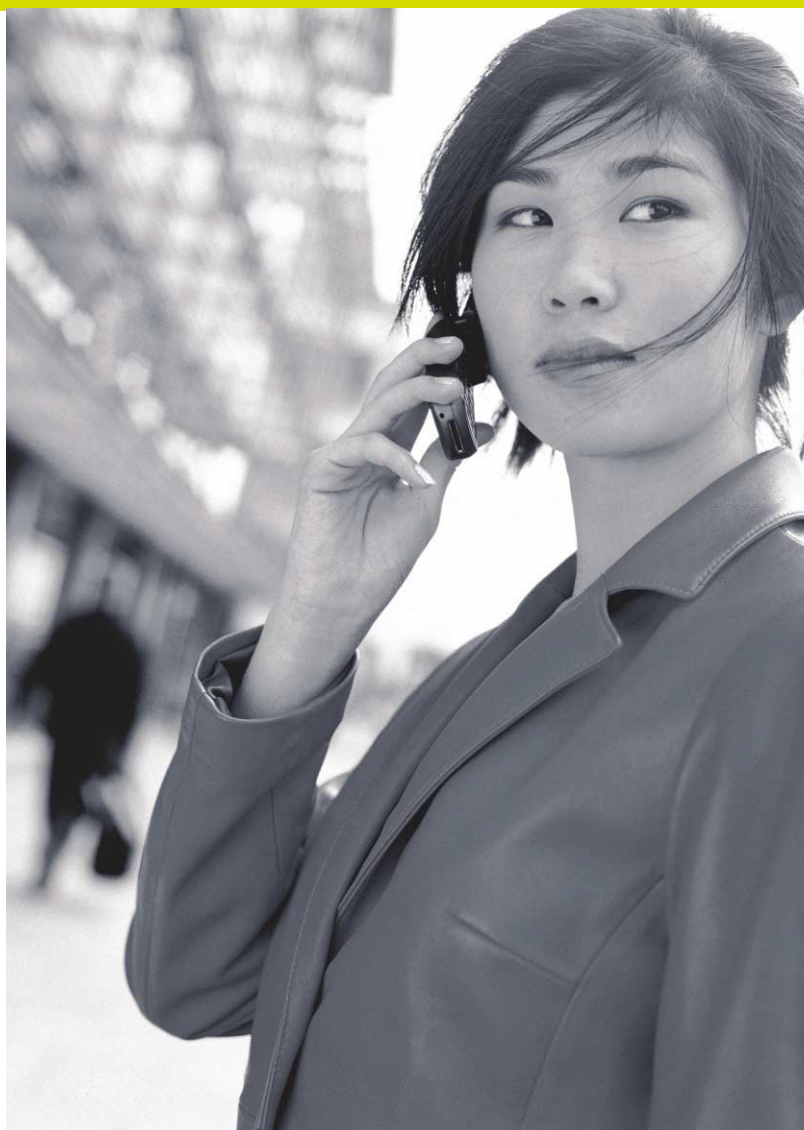




CEVA-X ARCHITECTURE CEVA-X1620



CEVA-X1620 Target Markets

- › 3G cellular handsets and Software Radio
- › SmartPhones / PDAs
- › Video & Audio processing for mobile devices
- › VoIP Gateways & broadband modems
- › Home entertainment (Digital TV, HDTV, PVR, DVD)

CEVA-X1620 is the first implementation of the CEVA-X DSP Cores family.

CEVA's Licensable DSP Cores

CEVA, Inc. is the world leading licensor of programmable Digital Signal Processors (DSP) Cores and integrated-applications to the semiconductor and electronics industry. CEVA's product line offers a variety of DSP cores. Each core delivers a different balance of performance, power dissipation and cost, allowing the customer to select a core preference in accordance to the targeted application requirements.

CEVA-X™ is a new architecture; the sixth generation of licensable DSP Cores in the company's portfolio of leading edge of DSP core technology solutions. It continues the line of the CEVA-Palm™, CEVA-Teak™, CEVA-TeakLite™, CEVA-Oak™ and CEVA-Pine™ DSP Cores.

CEVA-X DSP – Key Benefits

› High Performance at Low Power Consumption

The CEVA-X architecture has a unique mix of Very Long Instruction Word (VLIW) and Single Instruction Multiple Data (SIMD) architectures. The VLIW architecture allows a high level of concurrent instructions processing thus providing extended parallelism, as well as low power consumption. SIMD architecture allows single instructions to operate on multiple data elements resulting in code size reduction and increased performance. Low power consumption is also achieved in the CEVA-X by its instructions and dedicated mechanisms such as dynamic and selective units shutdowns and clock slow downs.

› Scalability

CEVA-X is a scalable architecture from which multiple DSP designs are derived. Each DSP design is aimed to serve market-specific needs that are characterized by performance, power consumption and cost. Its architecture offers a long-term performance roadmap that allows re-usability of legacy software and hardware platform designs. CEVA-X offers a scalable computation capability of 2, 4 and 8 Multiply-Accumulates (MACs). Furthermore, the scalability runs beyond the MACs – each MAC is part of computation cluster that includes additional logical, arithmetic and bit oriented mechanisms along with a dedicated register file. Using four such clusters results in a high performance architecture that executes multiple parallel operations in a single cycle, up to a peak performance of 11 billion instructions per second.

Additional scalability dimensions are: data word width that can be either 16 or 32 bits, to best meet markets requirements ranging from communications to home entertainment; customized size and type of data and program memories.

› Enables Customer Extensions

CEVA-X architecture enables customer differentiation via customer defined instructions and function extensions. Such extensions provide application and/or algorithms-specific added values, for example – video power-save extensions and advanced security. The extendibility is achieved while leveraging the benefits of a standard product with an open core framework, a third party community and a standard tool chain.

› High-level Programming

CEVA-X architecture enables efficient programming in C/C++ high-level languages that significantly reduces development cost and time-to-market.

The CEVA-X architecture was designed in conjunction with the CEVA-X compiler tool.

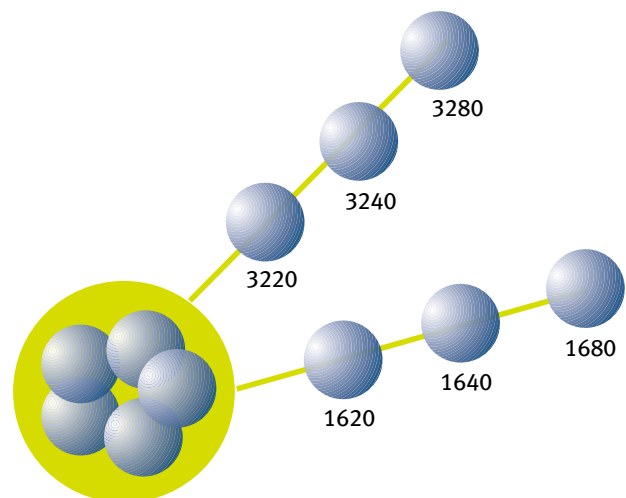
A most efficient and optimized C/C++ compiler is provided.

› Soft Core

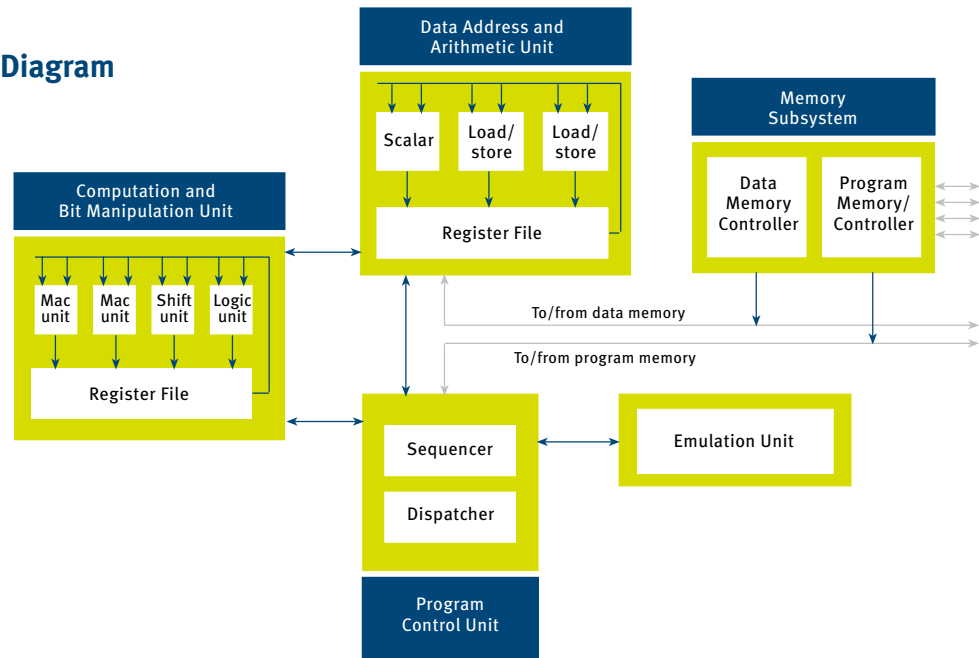
CEVA-X design implementations are Soft Core based, allowing the customer to select the optimal operating point in terms of die size, power consumption and performance. In addition, the customer has complete flexibility in selecting the foundry, process (e.g. 0.13μ, 90nm, 65nm) and complementary IPs.

CEVA-X IP incorporates fully automated design flow supporting mainstream EDA tools, significantly shortens time-to-market. CEVA-X design can be ported to an FPGA that can be used for product prototype, system integration, design acceleration and clarification

CEVA-X Architecture Roadmap



CEVA-X1620 Block Diagram



Development Tools, Software & Platforms

CEVA-X is supported by a complete set of software and hardware development tools. The software tools chain includes C/C++ Compiler, Assembler, Linker, Debugger, Simulator, Profiler, Utilities and DSP libraries; all under Integrated Development Environment (IDE). The hardware tools include various modular development system boards with associated accessories. A DSP hardware platform including the CEVA-X, DMA controller, CPU interfaces and a large offering of peripherals and interfaces is offered. Software and algorithms are provided by CEVA through its 3rd parties' network. CEVA-X DSP family – The first two digits from left represent the native data width, third digit represents the number of MAC units and the fourth digit represents the version.

CEVA-X1620 Advantages

- High frequency – up to 450 MHz @ 0.13μ worst case process
(Note: Frequency depends on configuration, synthesis, foundry, process, operating conditions, and libraries)
- Class-leading performance
 - Best Dual-MAC architecture
 - The CEVA-X1620 has the highest score that BDTI has

published for a licensable core as of Jan 2003*
Low power consumption achieved by the Core's instructions and a built-in mechanism, which shuts off any unused logic

- Enables Customer Extensions
- Highly efficient C Compiler
- Complete sets of software and hardware development tools

CEVA-X1620 Architecture Highlights

- Very Long Instruction Word (VLIW) and Single Instruction Multiple Data (SIMD) for enhancing DSP performance
- Load/store architecture
- Variable instruction width (16 or 32-bit) and variable length of instruction packets
- Concurrent execution of up to 8 instructions in parallel

* The BDTIsimMark2000 provides a summary measure of DSP speed. For more info and scores see www.BDTI.com. Score © 2004 BDTI.

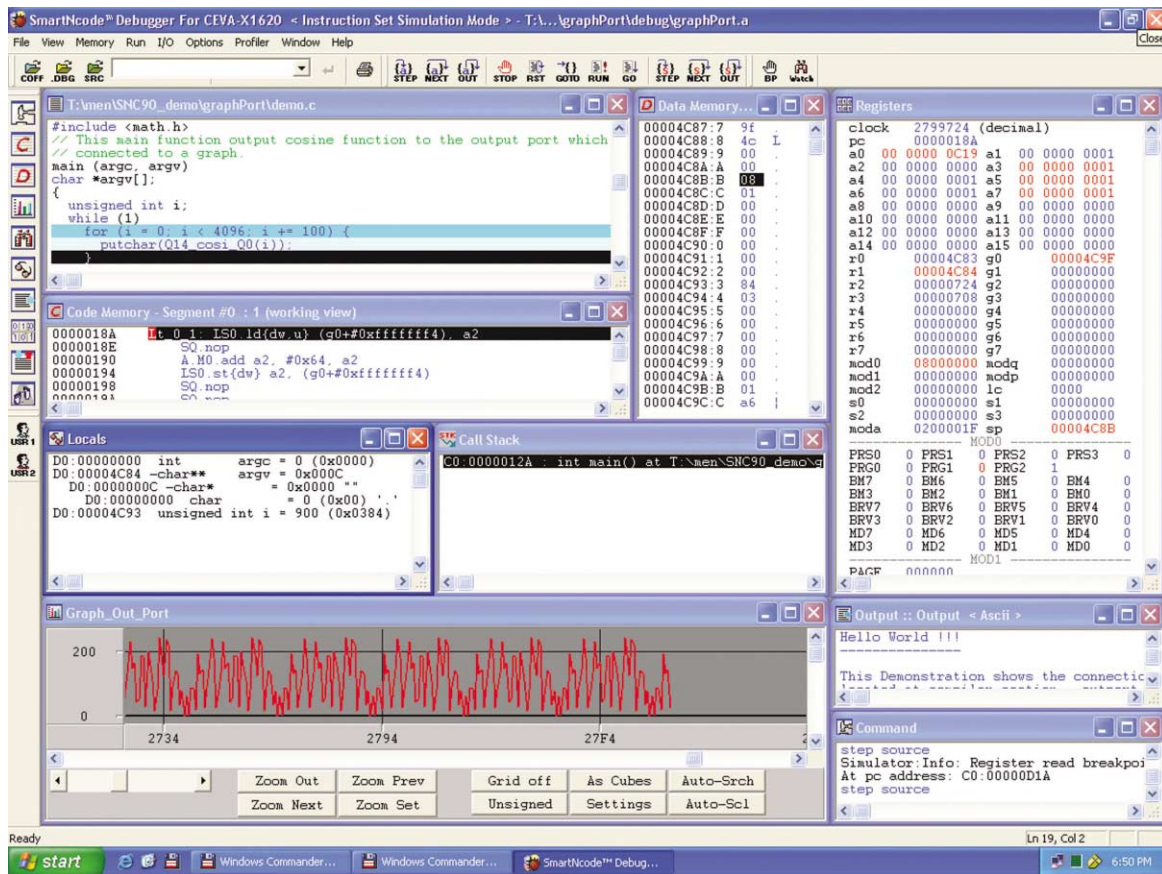
- All instructions can be predicated
 - Conditional execution
 - Reducing cycle count and code size on control and overhead code
- Computation unit contains
 - Two 16x16 bit two's complement MAC units, into 40-bit accumulators
 - Four 8x8 multiplications in single cycle
 - Two 16x16 multiplications in single cycle
 - Double precision multiplications support
 - Data scaling after multiplications or prior to storing to data memory
 - Unique MAC unit connectivity allowing fast and accurate filtering
 - Four 40-bit arithmetic units
 - 40-bit logical unit
 - 40-bit Bit Manipulation Unit, including full Barrel Shifter and Exponent unit
 - Two 40-bit data pack and unpack units
 - Sixteen 40-bit accumulators
 - Single cycle exponent evaluation of 40-bit values
 - Broad range of data byte operations
 - 2 cycle FFT butterfly
 - Single cycle Viterbi Add-compare-select
- Data Address Arithmetic Unit
 - Two Load-Store units providing up to 64-bit data bandwidth to the data memories
 - Eight 32-bit data memory pointers
 - 32-bit scalar unit for integer operations and fast arithmetic calculations on Address registers
 - Variety of data addressing modes (Indirect, Bit-reversed, Direct, Indexed, Stack)
 - Enables linear and cyclic/modulo registers post-modification
- Two Levels Memory
 - Up-to 4G-bytes address space
 - 64K-byte level 1 program space
 - 64K-byte level 1 data space
- Byte addressable data space
- Nine stages pipeline

- Interfaces
 - Separate I/O space for peripherals
 - AHB-Lite interfaces (Data, Program, External transfer request) for easy integration in a SoC
- Extensions – tightly coupled hardware extensions interface
- On-Chip Emulation Module (OCEM) support via JTAG port
- Memory Sub-system
 - Memory access arbitration
 - Controls AHB interface
 - Data DMA
 - Program DMA
 - Simple program cache controller

Software Development Tools

The CEVA-X1620 is accompanied by the advanced Integrated Development Environment (IDE) based Software Development Tools for embedded applications, supporting Windows and Solaris operating systems

- IDE Features
 - Project management
 - Makefile generation
 - Tools/project settings
 - Complete tools connectivity
 - Source control connectivity
 - Fully featured editor
 - Browsing information
 - Online help
- Highly optimizing C Compiler
 - High-end optimization exploiting the Core's architecture for efficient code generation
- Macro Assembler and Linker
 - Architectural restrictions checking and errors overcoming mechanism
 - Advanced mapping mechanism (Auto, Semi-auto and Manual)
 - Section overlay and multi-paging support



Debugger Screen View

- Advance Graphic User Interface Debugger
 - Instruction set and cycle accurate simulators
 - Emulation support (simulator's look & feel)
 - Source level debugging support – C and assembly files
 - User customizable DLLs for Debugger adaptation (Simulator extension, HW interface, etc.)
 - Parallel port, USB and JTAG emulation interfaces
 - Run-time violation detection by the simulator
 - Extensive breakpoint support, interrupt simulation and I/O support
 - Tight MATLAB Bi-Directional connectivity
 - Integrated graphic application Profiler
- Various Utilities and Converters
 - Libraries generation
 - Produce various memory burnable formats

Hardware Development Tools

Evaluation and Development Platform

The Evaluation and Development Platform (EDP) is used for CEVA-X1620 based application development.

The EDP interfaces to a host PC through the JTAG interface

(JBox) for full application control and contains the following:

- On-board fast SRAM
- On board 266MHz DDR SDRAM (without DIMM)
- On board Flash memory
- TFT LCD
- EDP interfaces
 - UART
 - Parallel port
 - Stereo audio in
 - MMC/SD flash socket
 - Analog video in/out
 - ARM core module connector (expansion slots)
 - USB2.0
 - 6 channels audio out
 - Passive microphone
 - CMOS sensor
 - Digital Video in/out

DSP on FPGA

A dedicated tool allows FPGA implementation of the CEVA-X DSP. The user can build an FPGA emulation model of the SoC containing the CEVA-X and other components, in order to verify the design before taping-out the actual silicon.



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